SUSTAINABLE DEVELOPMENT & GREEN BUILDINGS

Complete Question Bank: 2-Mark, 5-Mark & 10-Mark Questions (Simple English)

Overview & Exam Structure: This study guide contains exam questions and answers written in simple, clear English grounded strictly in your lecture sources:

- **2-Mark Questions:** Weeks 1, 2, and 4 (Core definitions and metrics)
- **5-Mark Questions:** Weeks 5 and 6 (Renewable energy & passive systems)
- **10-Mark Questions:** Week 3 (Air pollution, meteorology, acid rain, & indoor air quality)

PART A: 2-MARK QUESTIONS (WEEKS 1, 2 & 4)

Week 1: Sustainable Development & Resource Consumption

Q1: What are the three main pillars of sustainable development?

Answer: The three main pillars are **social**, **environmental**, and **economic sustainability**. Achieving sustainable development requires balancing all three pillars together.

Q2: Define 'Earth Overshoot Day' and state how it is calculated.

Answer: Earth Overshoot Day marks the date when humanity's annual demand on nature exceeds what Earth can regenerate in that year. It is calculated by comparing humanity's global ecological footprint with Earth's total biocapacity.

Q3: How does a 'sustainable building' differ from a 'green building' in terms of resource consumption?

Answer: A green building focuses mainly on saving energy, water, and materials during its daily operation. A sustainable building minimizes resource consumption across its entire life cycle—from extracting raw materials to building construction, operation, and eventual end-of-life disposal or reuse.

Week 2: Sustainable Urbanization & Green Buildings

Q4: What is the Urban Heat Island (UHI) effect?

Answer: The Urban Heat Island effect occurs when urban areas become significantly warmer than nearby rural areas because dense materials like concrete, asphalt, and brick absorb and store heat.

Q5: What distance defines 'locally sourced materials' in green buildings, and why are they used?

Answer: Locally sourced materials are extracted and manufactured within **800 km** of the building site. They are used to shorten transport distance, reduce greenhouse gas emissions, and lower the building's overall environmental impact.

Q6: State two key differences between building demolition and deconstruction.

Answer: Demolition uses heavy machinery to quickly knock down a building, creating large amounts of mixed waste sent to landfills. Deconstruction is the careful, step-by-step dismantling of a building specifically to save materials for reuse and recycling.

Week 4: Climate Change Foundations & Policy

Q7: Define 'Climate Forcings' and distinguish between natural and human-made forcings.

Answer: Climate forcings are external factors that alter Earth's radiative heating balance. **Natural forcings** include volcanic eruptions and solar irradiance changes, whereas **human-made (anthropogenic) forcings** include greenhouse gas emissions and land-use surface albedo changes.

Q8: What is the difference between climate change 'Mitigation' and 'Adaptation'?

Answer: Mitigation means taking actions to reduce greenhouse gas emissions or remove carbon dioxide from the atmosphere. **Adaptation** means adjusting buildings, infrastructure, and habits to minimize harm from climate changes that are already occurring.

Q9: What are Carbon Credits?

Answer: A carbon credit is a permit that allows an emitter to release one ton of carbon dioxide (or equivalent greenhouse gas). Emitters can buy or sell these credits under carbon pricing systems to stay within legal emission limits.

PART B: 5-MARK QUESTIONS (WEEKS 5 & 6)

Week 5: Solar Energy & Passive Systems

Q10: Explain the five core elements required for effective passive solar building design.

Answer: Passive solar design regulates indoor temperature without mechanical heating/cooling using five integrated elements:

1. **Aperture (Collector):** Large glass windows facing the equator (south in the northern hemisphere) to admit sunlight during winter.
2. **Absorber:** Dark, hard interior surfaces (such as concrete floors or brick walls) that directly catch incoming solar heat.
3. **Thermal Mass:** Dense materials behind the absorber that store absorbed heat during the day and release it slowly at night.
4. **Distribution:** Natural heat movement (conduction, convection, and radiation) that spreads stored warmth across rooms.
5. **Control:** Roof overhangs, shading devices, and vents used to block harsh summer sun and prevent overheating.

Q11: Compare Flat Plate Collectors (FPC) and Evacuated Tube Collectors (ETC) used in solar thermal systems.

Answer: Comparison between Flat Plate Collectors (FPC) and Evacuated Tube Collectors (ETC):

- **Flat Plate Collector (FPC):** Features fluid tubes attached to a dark absorber plate inside an insulated, glass-covered box. While simple and reliable, FPCs lose heat to surrounding air through convection and radiation.
- **Evacuated Tube Collector (ETC):** Consists of double-walled glass tubes with a vacuum sealed between them. The vacuum eliminates air convection, drastically reducing thermal losses.
- **Performance:** ETCs achieve significantly higher working temperatures and operate far more efficiently in cold, cloudy climates than FPCs.

Q12: Describe the structural design and working principle of a Trombe Wall.

Answer: Structure and working principle of a Trombe Wall:

- **Structure:** A Trombe wall is an indirect-gain solar system consisting of a thick (8 to 16 inches) dark-painted concrete or masonry wall placed directly behind an exterior glass sheet with a small air gap between them.
- **Working Principle:** Sunlight passes through the glass and heats the dark outer surface. Heat travels through the dense masonry wall very slowly (about 1 inch per hour), radiating warmth into the indoor room in the evening.
- **Air Vents:** Vents at the top and bottom of the wall allow warm air in the gap to circulate into the room by natural convection during sunny hours. The vents are closed at night to prevent reverse heat loss.

Week 6: Ocean & Geothermal Energy Systems

Q13: Describe Ocean Thermal Energy Conversion (OTEC), its key requirements, and its primary benefits.

Answer: Ocean Thermal Energy Conversion (OTEC) details:

- **Working Principle:** OTEC generates power by utilizing the temperature difference between warm ocean surface water and cold deep-ocean water (from 800 to 1000 meters depth).
- **Temperature Gradient:** Requires a steady temperature gap of at least 20°C, restricting suitable locations to tropical ocean regions.
- **Conversion Cycles:** Uses open-cycle (evaporating warm seawater in a vacuum) or closed-cycle systems (evaporating low-boiling fluids like ammonia).
- **Key Benefits:** Delivers continuous 24/7 baseload electricity (unlike intermittent solar/wind) and can produce fresh drinking water as a byproduct.

Q14: Explain how energy is harvested from Tides and Ocean Waves.

Answer: Technologies for harvesting Tidal and Wave energy:

- **Tidal Barrages:** Dams built across ocean estuaries that trap water at high tide and release it through hydro-turbines during low tide.
- **Tidal Stream Turbines:** Underwater turbines driven directly by fast-flowing ocean tidal currents.
- **Wave Energy Converters (WECs):** Harvest wave power using *Oscillating Water Columns* (rising waves compress air inside a chamber to drive a turbine) or *Surface Buoys* (up-and-down wave motion drives hydraulic generators).

Q15: Explain Geothermal Energy, Enhanced Geothermal Systems (EGS), and Geothermal Heat Pumps.

Answer: Geothermal energy technologies:

- **Geothermal Energy:** Thermal energy stored deep inside the Earth originating from primordial formation heat and natural decay of radioactive rock minerals.
- **Enhanced Geothermal Systems (EGS):** Human-made underground reservoirs created by injecting high-pressure water into deep, hot, dry rock to open fractures, allowing fluid to circulate and carry heat to the surface.
- **Geothermal Heat Pumps (GHPs):** Shallow ground systems that use the constant underground soil temperature to heat buildings in winter and cool them in summer.

PART C: 10-MARK QUESTIONS (WEEK 3)

Week 3: Air Pollution, Atmospheric Dispersion & Indoor Air Quality

Q16: Air Pollution Classification, Meteorology, and Plume Dispersion

(a) Air Pollutant Classification:

• *Primary vs. Secondary:* Primary pollutants are emitted directly from sources (e.g., CO, SO2, NO, primary soot). Secondary pollutants form in the atmosphere through chemical reactions between primary pollutants and air components (e.g., ground-level ozone, sulfuric acid, PANs).

• *Criteria vs. Non-Criteria:* Criteria pollutants are common contaminants with legally enforced health standards (CO, SO2, NO2, O3, lead, PM). Non-criteria pollutants are toxic or hazardous substances regulated based on specific industrial risks.

(b) Meteorological Influences & Temperature Inversion:

• *Meteorological Factors:* Wind speed, air temperature, and atmospheric stability dictate how quickly pollutants mix and dilute in the air.

• *Temperature Inversion:* Normally, warm surface air rises and disperses pollutants. During a temperature inversion, a layer of warm air sits above cooler surface air, trapping air pollutants near ground level.

(c) Plume Dispersion Behaviors:

• *Fanning:* Occurs under highly stable inversion conditions; the smoke plume spreads horizontally like a flat fan with no vertical mixing.

• *Lofting:* Occurs when a tall chimney releases smoke above an inversion layer; pollutants disperse upward safely away from the ground.

• *Fumigation:* Occurs when an inversion layer above breaks down from below; trapped pollutants are brought rapidly down to ground level in high concentrations.

Q17: Local & Regional Air Pollution: Photochemical Smog & Acid Deposition

(a) Industrial Smog vs. Photochemical Smog:

• *Industrial Smog:* Caused by burning coal/oil containing sulfur; consists of SO2, fog, and soot particles in cool, humid weather.

• *Photochemical Smog:* Caused when solar UV light triggers chemical reactions between vehicle emissions (NOx) and volatile organic compounds (VOCs); creates ground-level ozone and eye-irritating chemicals in warm, sunny weather.

(b) Acid Deposition & The Tall Smokestack Dilemma:

• *Formation:* SO2 and NOx emissions react with atmospheric moisture and oxygen to form sulfuric acid (H2SO4) and nitric acid (HNO3), which fall as acid rain, snow, or dry particles.

• *Tall Smokestack Dilemma:* Building tall smokestacks reduced local pollution, but pushed pollutants high into the air where winds carried them hundreds of kilometers away, creating regional acid rain across state borders.

(c) Mitigation Strategies:

• *Local Controls:* Installing catalytic converters on vehicles, using low-VOC paints, and implementing fuel vapor recovery.

• *Regional Controls:* Equipping power plants with flue gas scrubbers for SO2 and selective catalytic reduction (SCR) for NOx.

Q18: Global Air Quality Issues & Indoor Environmental Quality (IAQ)

(a) Stratospheric Ozone Depletion & Antarctic Ozone Hole:

• *Ozone Chemistry:* UV light splits oxygen molecules (O2) into O atoms, which combine with O2 to form ozone (O3). Synthetic chemicals like CFCs release chlorine (Cl) radicals under UV light.

• *Catalytic Cycle:* Cl + O3 → ClO + O2, and ClO + O → Cl + O2. One chlorine atom acts as an unconsumed catalyst, destroying up to 100,000 ozone molecules.

• *Antarctic Ozone Hole:* Extreme winter cold forms Polar Stratospheric Clouds (PSCs). In spring (Sept–Oct), returning sunlight triggers rapid catalytic destruction, causing ozone levels to drop below 220 Dobson Units.

(b) Ozone-Depleting Substances (ODSs) & Montreal Protocol:

• *ODSs:* Chlorofluorocarbons (CFCs), halons (fire extinguishers), and methyl bromide (agricultural fumigant).

• *Montreal Protocol (1987):* A universally ratified international treaty that phased out ODS production, enabling the gradual recovery of the stratospheric ozone layer.

(c) Indoor Air Pollution (IAP) & Management:

• *Severity:* Indoor air is often 2 to 5 times more polluted than outdoor air, causing Sick Building Syndrome (SBS).

• *Four Most Dangerous Indoor Pollutants:*

1. **Tobacco Smoke:** Carcinogenic particles causing lung and heart disease.
2. **Formaldehyde:** Toxic gas offgassed from pressed-wood furniture, carpets, and glues.
3. **Radon-222:** Radioactive gas seeping into basements from soil/rock, causing lung cancer.
4. **Particulate Matter:** Fine soot/dust from unvented cooking and heating stoves.

• *IAQ Management:* **Source Control** (using low-VOC materials and banning smoking), **Ventilation** (increasing fresh outdoor air exchange), and **Air Filtration** (using HEPA air purifiers).